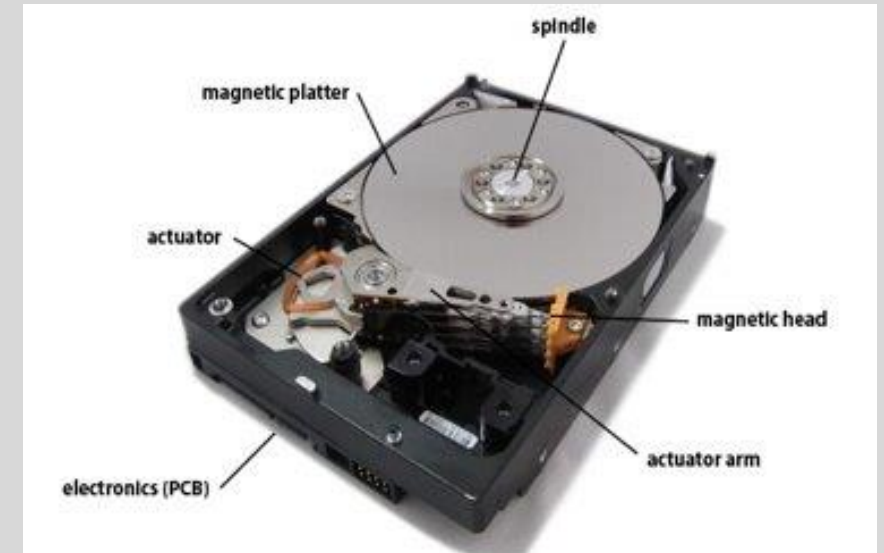




SECONDARY STORAGE

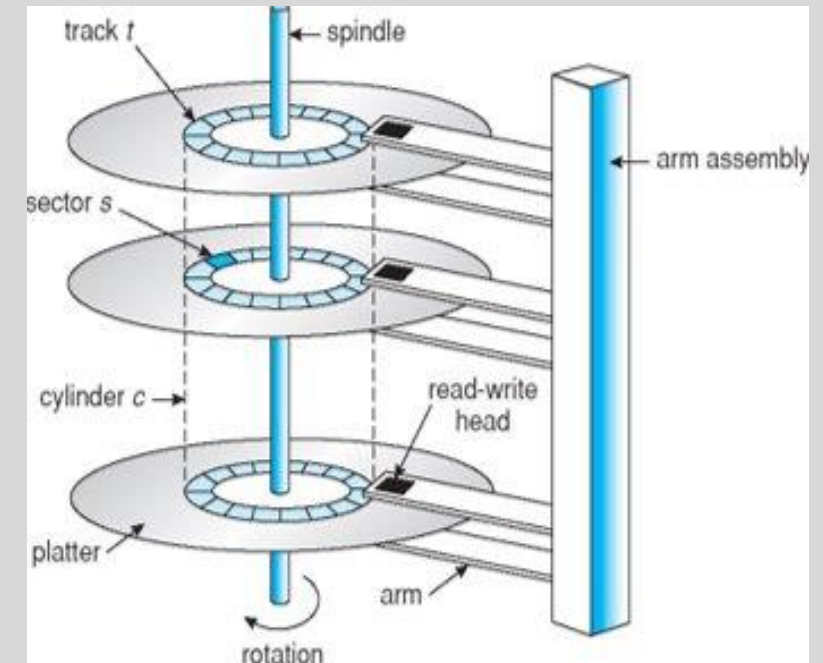
Hard Disk Drive

- Still widely used for large-scale data storage (multi-terabyte to petabyte scale) due to cost-efficiency.
- **Electromechanical Components**
 - **Rotating Platters**
 - Multi-platter setups for higher capacity
 - Helium-filled drives reduce friction, increase platter count
 - **Actuator Arm Assembly**
 - Supports **dual-actuator technology** (e.g., Seagate's MACH.2) for faster parallel access
- **Electronic & Intelligent Components**
 - **Disk Controller**
 - Coordinates read/write operations
 - Now includes **AI-based predictive failure detection** and **error correction (ECC)**
 - **Cache / Buffer Memory**
 - DRAM-based buffer (often 256MB to 1GB)
 - **Intelligent prefetching** for sequential and random access optimization
 - **Interface Controller**
 - Supports modern interfaces like **SATA III**, **SAS**, and **NVMe over PCIe**
 - Enables integration with hybrid systems (HDD + SSD tiers)



Hard Disk Drive: Structure and Organization

- A **Hard Disk Drive** consists of one or more **spinning magnetic platters** used to store data persistently.
- A movable **read/write head** floats just above the surface of each platter, accessing data while the disk spins.
- Data is organized in **concentric circles** on the disk surface, known as **tracks**.
- Each track is further divided into smaller units called **sectors**.
- A typical sector stores **512 bytes** of data, although modern drives may support **4KB sectors** (Advanced Format).
- Multiple platters and double-sided surfaces allow for **very high storage density**.



Hard Disk Drive (HDD): Physical Characteristics

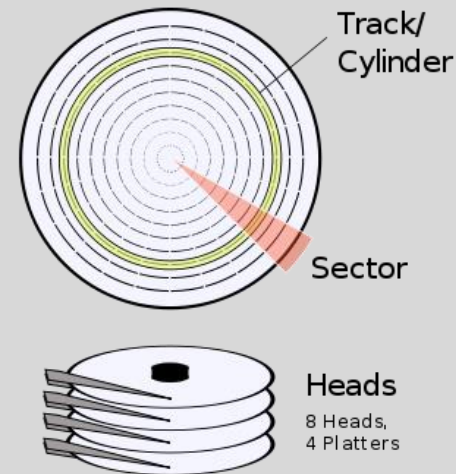
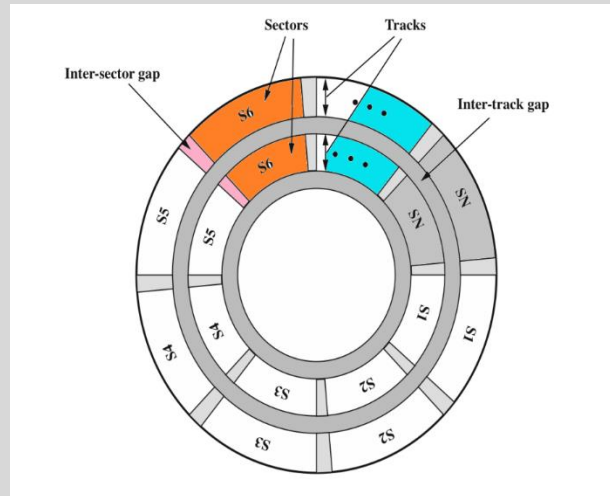
- **Platter Sizes:** Common diameters include **3.7"**, **3.3"**, and **2.6"**
- **Rotation Speeds (RPM):**
 - Typical values: **5400, 7200, 10000, 15000**
 - Minor variation: around **0.5% to 1%** due to mechanical tolerance
- **Number of Platters:** Usually between **1 to 5**
- **Power Consumption:**

Proportional to

$$\text{Power} \propto (\text{Number of Platters}) \times (\text{RPM})^{2.8} \times (\text{Diameter})^{4.6}$$
- **Read/Write Head:**
 - A **tiny electromagnetic component** positioned above the platter surface
 - **Reads** magnetic fields and **writes** data by altering them
 - Mounted on an actuator arm that moves across tracks with high precision

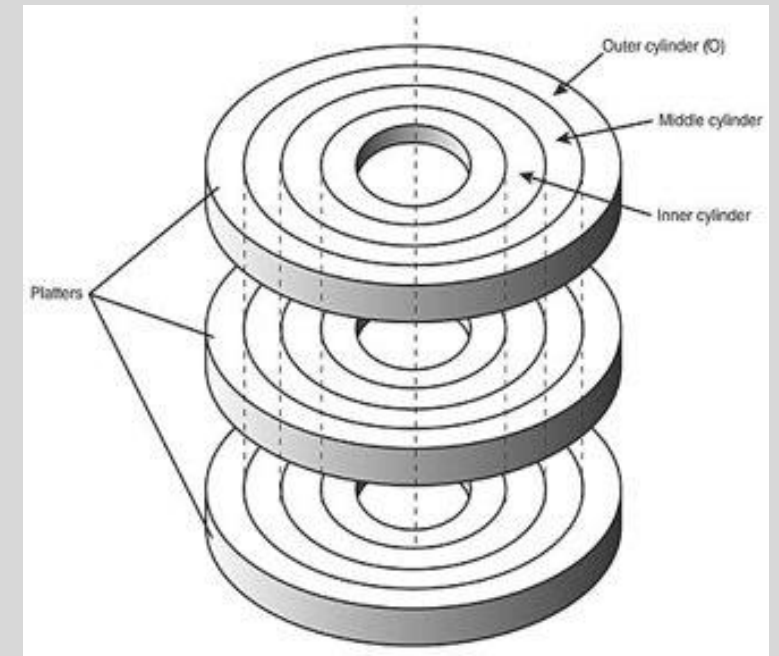
Hard Disk Platter and Head Structure

- Each surface of a platter is accessed by a read/write head. A single platter has two heads, one on the top and one on the bottom.
- Hard disk drives can have multiple platters stacked together to increase storage capacity.
- Each head reads and writes data using concentric circular paths called tracks.
- Tracks are divided into smaller units called sectors. Sectors are the smallest addressable storage units, typically 512 bytes or 4 KB.



Cylinder in Hard Disk Drives

- A cylinder is the set of all tracks that lie directly above and below each other on every platter surface at a specific arm position.
- It forms a three-dimensional collection of tracks, one track per platter surface, all aligned vertically.
- When the disk arm moves to a cylinder, the heads can read/write data on all tracks of that cylinder without additional arm movement, minimizing seek time.
- Cylinders help organize data physically, enabling disk scheduling algorithms to optimize read/write by grouping data access within the same cylinder.
- Addressing Scheme: Disk addresses often include cylinder number, head (surface), and sector, simplifying the mapping of logical blocks to physical locations.



Logical Block Addressing in Disk Drives

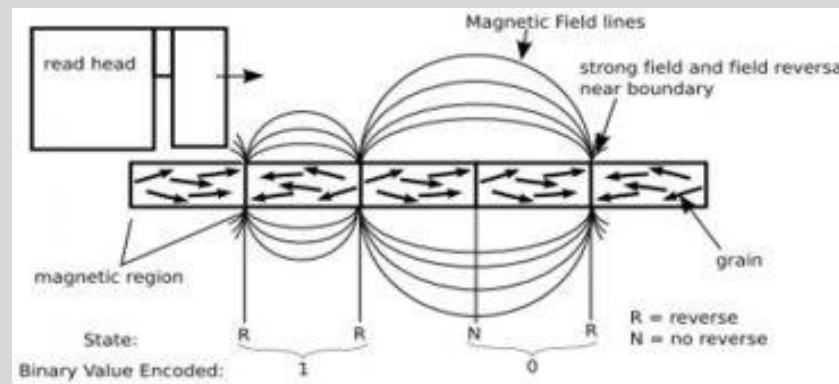
- Disk drives are organized as large one-dimensional arrays of logical blocks, with the logical block being the smallest unit of data transfer.
- Low-level formatting defines these logical blocks on the physical media.
- The sequence of logical blocks maps directly and sequentially onto the disk's physical sectors.
- Logical block 0 corresponds to the first sector of the first track on the outermost cylinder.
- The mapping progresses sequentially through the sectors of that track, then continues through all tracks within the cylinder, and finally moves inward through the cylinders from outermost to innermost.

Example: Logical Block Addressing on a Disk

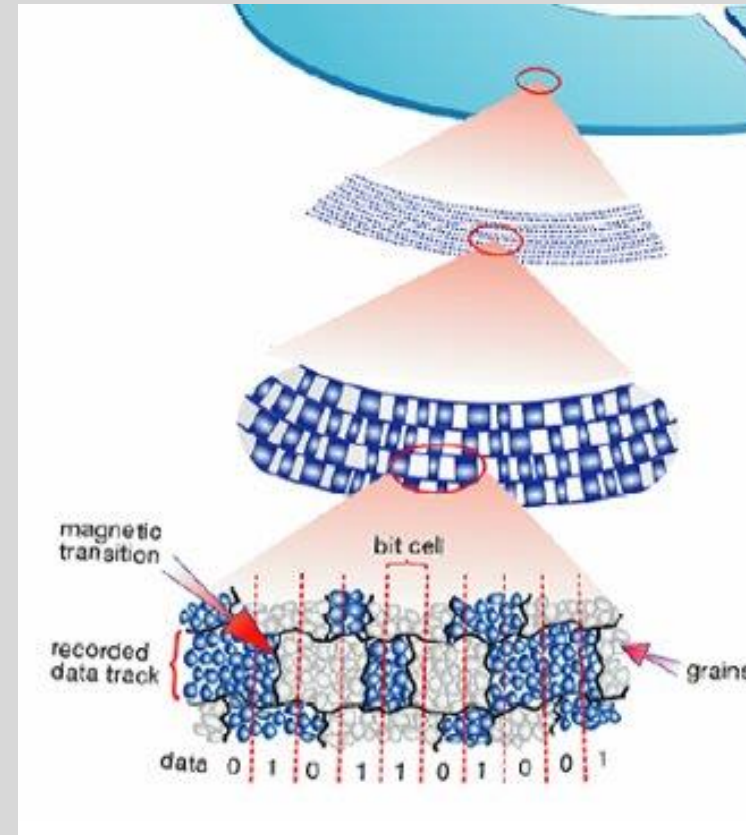
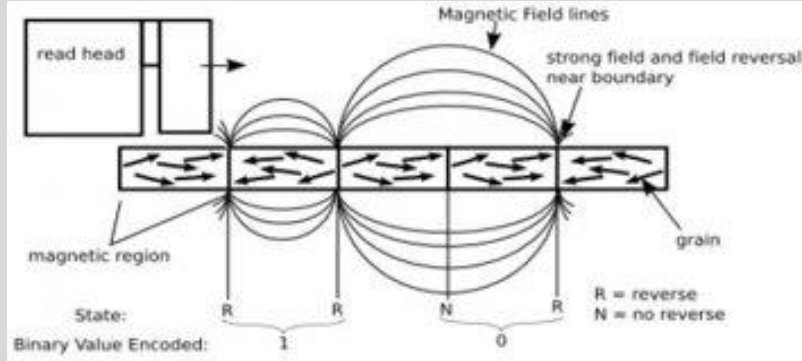
- Suppose a disk has:
 - **100 cylinders**
 - **10 tracks per cylinder** (each surface has 10 tracks)
 - **20 sectors per track**
- **Logical block 0** corresponds to:
 - Cylinder 0 (outermost)
 - Track 0 (first track on that cylinder)
 - Sector 0 (first sector on that track)
- **Logical block 1** is:
 - Cylinder 0, Track 0, Sector 1
- This mapping continues through all sectors of the first track, then moves to:
 - Cylinder 0, Track 1, Sector 0
 - Cylinder 0, Track 1, Sector 1
 - ... and so forth until the last track of cylinder 0
- After finishing cylinder 0, it moves to cylinder 1 starting again at:
 - Track 0, Sector 0
- Thus, logical blocks are sequentially mapped onto physical sectors progressing from outer to inner cylinders.

Bit Cell in Magnetic Storage

- A **bit cell** is made up of many tiny magnetic grains, typically between **50 and 100 grains per bit**.
- Each grain is extremely small, on the order of **10 nanometers** in size.
- A **'0' bit** is represented by a region where all grains have the **same magnetic polarity** (uniform magnetization).
- A **'1' bit** is represented by the boundary between two regions with **opposite magnetic polarities** (a transition in magnetization).
- The **read/write head** detects these magnetic polarities as it passes over the disk and can also change the magnetization to write data.



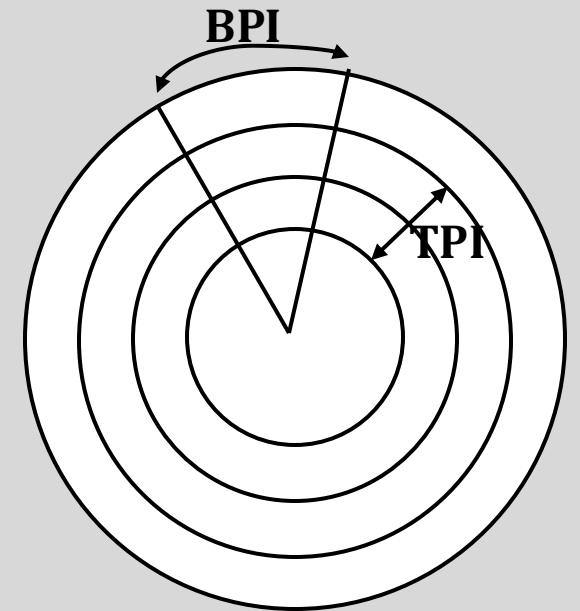
Bit 0 and Bit 1



Storage Density

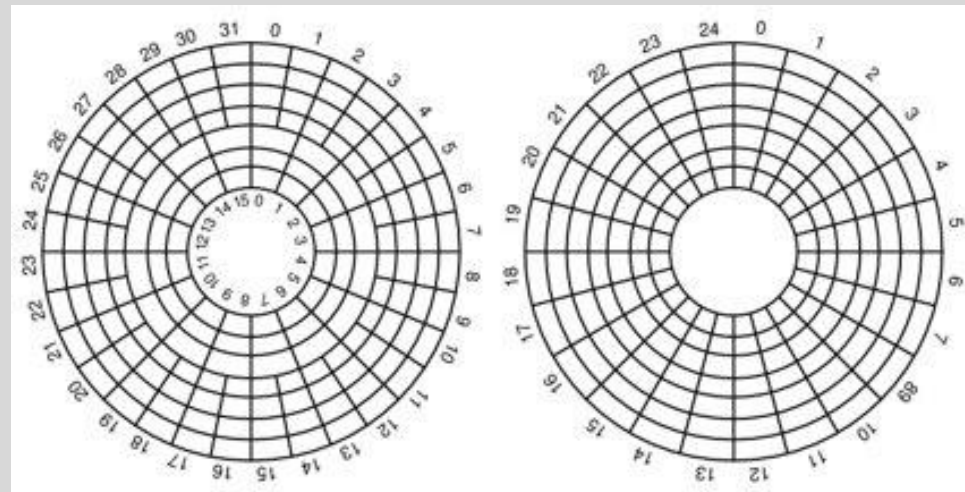
- Determines both **capacity** and **performance** of a storage device
- **Density Metrics:**
 - **Linear Density:** Number of bits per inch along a track (Bits Per Inch, BPI)
 - **Track Density:** Number of tracks per inch across the disk surface (Tracks Per Inch, TPI)
 - **Areal Density:** The product of linear density and track density

$$\text{Areal Density} = \text{BPI} \times \text{TPI}$$



HDD-Bit Density

- Constant Linear Velocity (CLV):
 - The bit density per track decreases on the outer tracks to maintain a constant linear speed of the disk surface under the read/write head.
- Constant Angular Velocity (CAV):
 - The disk spins at a fixed rotational speed (constant RPM).
 - Outer tracks, having larger circumference, hold more sectors per track than inner tracks.
 - The number of sectors per track is fixed during formatting and does not change dynamically.
 - This allows uniform rotation speed but variable data density, with more data on outer tracks.



Disk Access Time

- **Definition:**

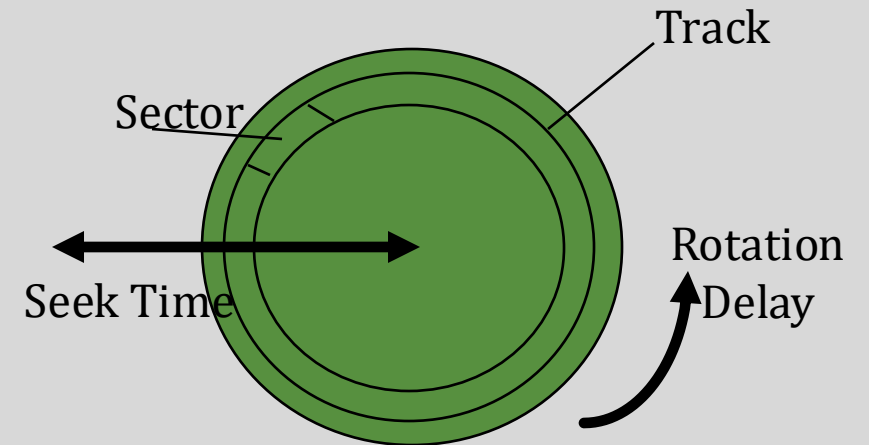
Disk access time is the total time required to read or write data from/to a disk. It includes the time to position the read/write head and transfer the requested data to memory.

- **To read from the disk, the following parameters must be specified:**

- Cylinder number (which track)
- Surface number (which platter face)
- Sector number (which block on the track)
- Size of data to transfer
- Memory address where data will be stored

- **Disk access time consists of three main components:**

- **Seek Time:** Time taken to move the read/write head to the correct track (cylinder)
- **Rotational Latency:** Time waiting for the desired sector to rotate under the head
- **Transfer Time:** Time to actually transfer the data from the disk to memory



Seek Time

- **Definition:**

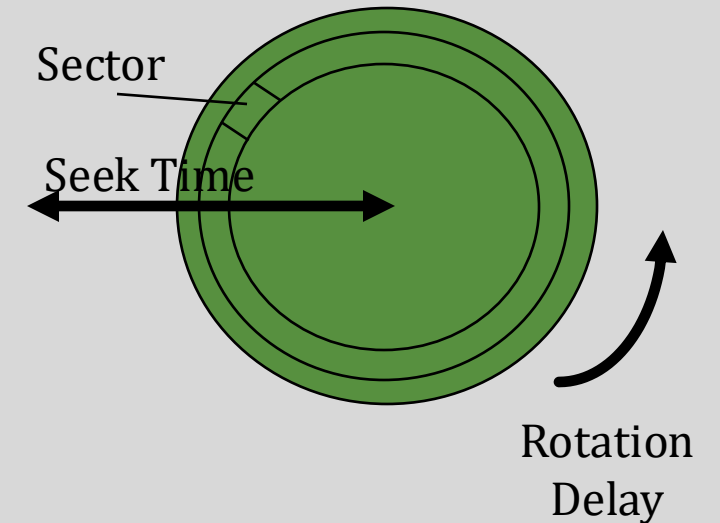
Seek time is the time required for the disk's read/write arm to move the head to the correct track on the platter.

- **Seek time depends on several factors:**

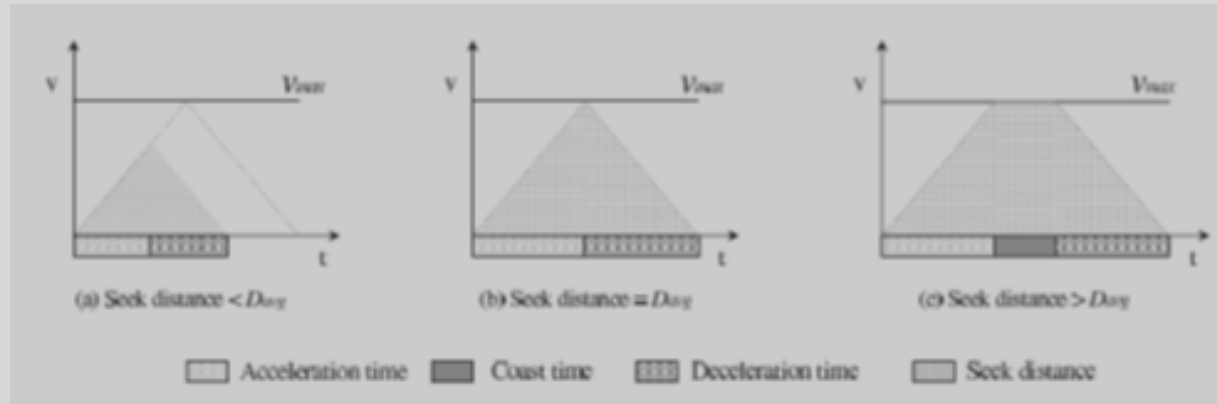
- The inertial power of the arm actuator motor (how quickly it can accelerate and decelerate)
- The distance between the outer and inner recording radii of the disk (also called the data band)
- The physical size of the platter

- **Components of a seek operation:**

- **Speedup:** The arm accelerates from rest
- **Coast:** The arm moves at maximum velocity (especially during longer seeks)
- **Slowdown:** The arm decelerates as it approaches the target track
- **Settle:** Fine adjustments to position the head precisely over the desired track



Variation in Seek Time



- Very short seeks (2–4 cylinders):
Settle time dominates because the arm only needs fine adjustment to position the head precisely.
- Short seeks (100–200 cylinders):
Speedup and slowdown phases dominate as the arm accelerates and decelerates quickly over the short distance.
- Longer seeks (> 200 cylinders):
Coast time dominates since the arm moves at maximum velocity for the majority of the seek distance.
- Impact of smaller platter sizes and higher track densities (TPI):
Settle time becomes increasingly important as precision in head positioning is critical for accurate data access.

Disk Formatting

- Low Level Formatting
- High Level Formatting

Low-Level Formatting (LLF)

- Also known as **physical formatting**.
- Defines the physical structure of the disk by marking the locations of **tracks and sectors** on the platter.
- Writes **control data** to identify the start of each sector and ensure proper timing/synchronization during read/write operations.
- Modern hard disks are **low-level formatted at the factory**, and this process is **not typically user-configurable** today.
- LLF also includes error correction structures like ECC (Error Correcting Code) at the sector level.

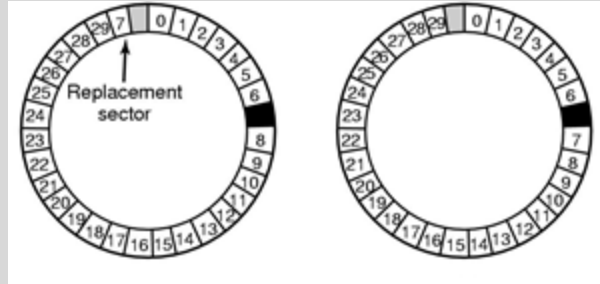
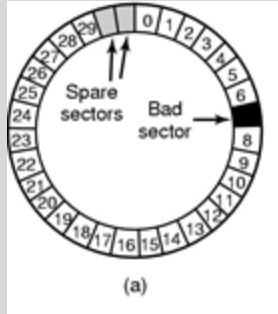
High-Level Formatting (HLF)

- Also known as **logical formatting**.
- Prepares the disk for use by an **operating system**.
- Creates **file system structures** such as:
 - Master Boot Record (MBR)
 - File Allocation Tables (FAT) or other directory structures (e.g., inode tables in ext4)
- Initializes partitions and metadata necessary for managing files and folders.
- Typically done when:
 - Installing or reinstalling the OS.
 - Reinitializing the disk for a fresh start.
 - Changing file systems (e.g., from NTFS to ext4).

Bad Sector in Disk Management

- A **bad sector** is a portion of the disk that becomes **unreadable** or **unwritable** due to physical damage or media degradation.
- Bad sectors are typically identified during:
 - **Low-level formatting (LLF)** at the factory, or
 - By system utilities like **CHKDSK**, **SCANDISK**, or during disk I/O operations.
- These sectors are **marked as unusable** and are excluded from logical-to-physical block mapping.
- If a file is assigned to a sector later found to be bad:
 - The disk controller or OS **remaps** that logical block to a **spare sector** reserved for this purpose.
 - This process is **transparent** to higher-level software and maintains **sequential logical block addressing (LBA)**.
- As a result, although the logical block number remains consistent, the underlying physical location **changes silently**, helping preserve data integrity.

Techniques of Bad Sector Management



Sector forwarding Sector slipping

- Sector Forwarding (also called **replacement** or **remapping**) is when the disk controller **redirects access** to a bad sector to a **spare sector** somewhere else on the disk.

Example: Sector 7 is remapped after sector 29

- Sector Slipping is a **proactive** remapping strategy. It **shifts the data** on the disk to maintain sequential physical order **after detecting a bad sector**.

Example: sector 7 onwards is shift place as shown in the diagram

Overview of Nonvolatile Memory Devices (NVM)

- What is NVM?
- Flash-based storage without moving parts
- Can appear as:
 - Solid State Drives (SSDs) (like 2.5"/3.5" form factors)
 - USB flash drives / thumb drives
 - DRAM-style chips
 - Onboard storage in smartphones/laptops



Figure 11.3 A 3.5-inch SSD circuit board.

Pros and Cons of NVM

- Advantages of NVM

- **No seek time** or **rotational latency**
- **More reliable** (no moving parts)
- **Lower power consumption**
- Increasing **capacity** and **falling prices**
- Common in laptops for speed & portability

- Limitations of NVM

- **Higher cost** per MB than HDD
- **Lower capacity** (historically)
- **Limited write endurance** (wear-out after many erases)

NAND FLASH Characteristics

- **Read:** Fastest operation
- **Write:** Slower than read
- **Erase:** Slowest; must erase before overwrite
- Erase happens **block-wise**, write/read **page-wise**
- NAND cells **wear out** after ~100,000 erase cycles
- Lifespan & Wear Metrics
 - Measured in **Drive Writes Per Day (DWPD)**
 - Ex: 1TB drive with 5 DWPD → can handle 5TB writes/day
 - Wear leveling & error correction done by **controller**
- System Integration
- Some SSDs connect via **PCIe**, bypassing slow buses
- Used as:
 - Disk drive replacements
 - Cache tiers between HDD and RAM

OS View and Device Logic

- OS reads/writes **logical blocks**
- NVM device controller handles:
 - **Erase-before-write**
 - **Wear-leveling**
 - **Parallelism** across NAND dies
 - **Bad block management**

NAND Flash Controller Algorithms

- **NAND flash cannot overwrite** existing data, it must **erase first**.
- **Pages** hold data; **blocks** are collections of pages.
- A written page becomes **invalid** when its data is outdated.
- Erase operations are performed at the **block** level, not page level.

Flash Translation Layer (FTL)

- FTL is a mapping table inside the SSD controller.
- Maps **logical blocks** → **physical pages**.
- Keeps track of:
 - Which pages are **valid** or **invalid**.
 - Which blocks are **ready to be erased**.
 - Think of FTL as the file system for the SSD's internal memory.

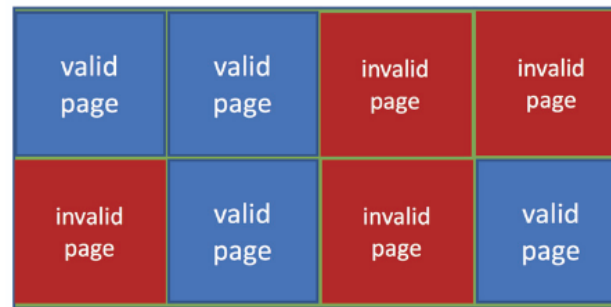


Figure 11.4 A NAND block with valid and invalid pages.

Garbage Collection

- If SSD is full but many pages are invalid, it triggers **garbage collection**.
- Valid data from mixed blocks is:
 - **Copied to new blocks**
 - Old blocks are then **erased**
- Frees up space for new writes.
- Over-Provisioning
 - A portion of SSD (often ~20%) is reserved.
 - Used to:
 - Temporarily hold data during garbage collection
 - Ensure there's always space to write
 - **Improves write performance and prevents failure when full**

Wear Leveling and Data Protection

- Flash memory **wears out** with each erase cycle (~100,000 cycles).
- Wear leveling distributes write/erase cycles **evenly across blocks**.
- Prevents early failure of frequently used blocks.
- Prolongs overall SSD lifespan.
- Uses **Error-Correcting Codes (ECC)**:
 - Detects and corrects small errors in flash memory.
 - Pages with frequent errors may be marked as **bad**.
- Supports RAID-like protection for **catastrophic failures**.



Thank You